

Moving-Hole Defects in Translated Reduced-Residue BDH Variances:

A Rank-Two Formula and Critical Block Bound

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Abstract

Reduced-residue Barban–Davenport–Halberstam (BDH) variances distinguish the zero residue. Translating a physical interval therefore replaces the deleted zero residue by a modulus-dependent moving hole. We isolate and prove the exact correction. Deleting a residue is an orthogonal rank-one loss from the all-residue variance; changing the deleted residue is consequently rank two, with exact nonzero spectrum $\pm\sqrt{q(q-2)}/(q-1)$. We lift the identity through the mandatory $(q-2)/(q-1)$ diagonal subtraction and through four-packet complex polarization. For length- $O(H)$ bounded-overlap blocks, moduli $q \in (Q, 2Q]$, and a Schwartz localized kernel, we then prove the deterministic bound

$$\sum_{b,c} |M_{bc}| \ll A_\beta A_w J(H^2 + HQ + Q^2).$$

At the frozen scales $H = x^{21/32}$, $Q = x^{1/3}$ and $J = x/H x^{o(1)}$, this is $x^{53/32+o(1)} = x^{5/3-1/96+o(1)}$. Thus the translation subgate, and only that subgate, pays the critical $1/96$ clock and the required strict margin for every fixed $1/400 < \delta' < 1/96$. The claim is structural L1: we do not prove the prime-only, q -weighted, kernel-localized signed zero-hole BDH theorem, a full Gate-B estimate, an arithmetic advance, an L^2 estimate, fixed-atom credit, or a twin-prime theorem.

1 Introduction and claim firewall

The V59 prime-dynamics reduction expresses one signed endpoint scalar as a four-packet polarization of reduced-residue BDH remainders [2]. A direct comparison with general-sequence BDH results is not translation invariant: after writing a physical variable as $n = s + m$, the condition $q \nmid n$ deletes the local residue $m \equiv -s \pmod{q}$, not the local zero residue. The purpose of this paper is to remove that translation mismatch exactly and to determine its full deterministic cost.

The answer has two parts. First, a finite-dimensional projector identity shows that the variance correction is rank two. The exact diagonal-subtracted remainder additionally requires a two-residue energy correction. Second, kernel localization and block geometry, rather than finite rank, yield the needed power: the correction reassembles with J effective blocks, not J^2 .

The status language throughout is

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V60_ROUTE_ADVANCE = YES,
V60_TRANSLATION_SUBGATE_DELTA = 1_OVER_96_PROVED,
V60_TRANSLATION_SUBGATE_STRICT_1_OVER_400 = PAID,
V60_FULL_GATE_B_STRICT_1_OVER_400 = UNPAID,
V60_ARITHMETIC_ADVANCE = NO,
V60_FIXED_ATOM_CREDIT = 0,
V60_L2 = NONE,      TPC_207_TRIGGER = true.
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No statement below is a twin-prime theorem. We use “prove” for the exact algebra and deterministic estimates established here; source comparisons are explicitly separated.

2 Reduced-residue BDH rows

Fix $q \geq 2$ and a complex row $z = (z_r)_{r \in \mathbb{Z}/q\mathbb{Z}}$. Put

$$\mu = \frac{1}{q} \sum_{r \in \mathbb{Z}/q\mathbb{Z}} z_r, \quad V_{\text{all}}(z) = \sum_{r \in \mathbb{Z}/q\mathbb{Z}} |z_r - \mu|^2.$$

For a deleted residue, or *hole*, $h \in \mathbb{Z}/q\mathbb{Z}$, define

$$\mu_h = \frac{1}{q-1} \sum_{r \neq h} z_r, \quad V_h(z) = \sum_{r \neq h} |z_r - \mu_h|^2.$$

For a prime modulus, the gcd-grouped variance used by Harper [1] has a singleton non-unit class and hence zero contribution from that class. Its prime row is therefore exactly V_0 . The translated physical row will instead be V_{h_q} for a modulus-dependent h_q .

We also require a diagonal ledger. Let $E_r \geq 0$ denote the coefficient energy in residue r , and set

$$\kappa_q = \frac{q-2}{q-1}, \quad R_h(z, E) = V_h(z) - \kappa_q \sum_{r \neq h} E_r. \quad (1)$$

The factor $q-2$ records the exact number of nonprincipal characters after the reduced-residue character expansion; replacing it by $q-1$ changes the scalar.

3 The moving-hole projector theorem

Theorem 3.1 (Moving-hole identity). *For every $q \geq 2$, $z \in \mathbb{C}^q$, and $h \in \mathbb{Z}/q\mathbb{Z}$,*

$$\boxed{V_h(z) = V_{\text{all}}(z) - \frac{q}{q-1} |z_h - \mu|^2.} \quad (2)$$

Consequently,

$$V_h(z) - V_0(z) = \frac{q}{q-1} \left(|z_0 - \mu|^2 - |z_h - \mu|^2 \right). \quad (3)$$

Proof. Write $u_r = z_r - \mu$. Then $\sum_r u_r = 0$ and

$$\mu_h = \mu - \frac{u_h}{q-1}.$$

For $r \neq h$ we have $z_r - \mu_h = u_r + u_h/(q-1)$. Since $\sum_{r \neq h} u_r = -u_h$,

$$\begin{aligned} V_h(z) &= \sum_{r \neq h} \left| u_r + \frac{u_h}{q-1} \right|^2 \\ &= \sum_{r \neq h} |u_r|^2 - \frac{2}{q-1} |u_h|^2 + \frac{1}{q-1} |u_h|^2 \\ &= V_{\text{all}}(z) - \frac{q}{q-1} |u_h|^2. \end{aligned}$$

Subtracting the cases h and 0 gives (3). □

Let $P = I - q^{-1} \mathbf{1} \mathbf{1}^*$ be the all-row centering projector and define

$$v_h = \sqrt{\frac{q}{q-1}} \left(e_h - \frac{1}{q} \mathbf{1} \right).$$

Then $\|v_h\| = 1$, $v_h \perp \mathbf{1}$, and Theorem 3.1 becomes

$$V_h(z) = \langle z, Pz \rangle - |\langle z, v_h \rangle|^2. \quad (4)$$

4 Sharp rank-two spectrum and obstruction

Proposition 4.1 (Exact spectrum). *Suppose $h \neq 0$ and $q > 2$. The self-adjoint translation-defect operator*

$$T_h = P_{v_0} - P_{v_h}, \quad P_v = vv^*,$$

has rank two and nonzero spectrum

$$\boxed{\text{spec}_{\neq 0}(T_h) = \left\{ \frac{\sqrt{q(q-2)}}{q-1}, -\frac{\sqrt{q(q-2)}}{q-1} \right\}.} \quad (5)$$

For $h = 0$ or $q = 2$, the defect is zero.

Proof. For $h \neq 0$,

$$\langle v_0, v_h \rangle = -\frac{1}{q-1}.$$

The difference of two rank-one orthogonal projectors has trace zero and is supported on $\text{span}(v_0, v_h)$. Its two eigenvalues are $\pm \sqrt{1 - |\langle v_0, v_h \rangle|^2}$, which gives (5). If $q = 2$, the two centered unit vectors differ only by sign, hence their projectors coincide. □

In particular,

$$\|T_h\| = \frac{\sqrt{q(q-2)}}{q-1} \longrightarrow 1.$$

Thus finite rank gives no asymptotic saving. This is a sharp obstruction to an argument that merely calls the correction “low rank.” A related coefficient-level obstruction is obtained by placing L equal coefficients A in a single residue: when the hole is moved across that residue, the diagonal-subtracted defect can have size

$$\frac{q-2}{q-1} L(L-1) |A|^2.$$

The block theorem below must therefore use residue counts, kernel decay, and collective geometry.

5 Exact diagonal-subtracted lift

Theorem 5.1 (Mandatory energy correction). *For the remainder (1),*

$$\boxed{R_h - R_0 = \frac{q}{q-1} \left(|z_0 - \mu|^2 - |z_h - \mu|^2 \right) + \frac{q-2}{q-1} (E_h - E_0).} \quad (6)$$

If the ambient BDH row carries outer weight q , then every term on the right of (6) must also be multiplied by q .

Proof. The variance difference is (3). The diagonal difference is

$$-\kappa_q \sum_{r \neq h} E_r + \kappa_q \sum_{r \neq 0} E_r = \kappa_q (E_h - E_0).$$

Adding the two identities proves the claim. $\square$

The energy correction is essential. For $q = 5$, take residue-0 coefficients 5, 5, residue-1 coefficient 1, and no others. Then

$$z = (10, 1, 0, 0, 0), \quad E = (50, 1, 0, 0, 0),$$

and exact calculation gives

$$R_0 = 0, \quad R_1 = \frac{75}{2}. \quad (7)$$

By contrast, the one-coefficient realization $E = (100, 1, 0, 0, 0)$ has $R_0 = R_1 = 0$; the row z alone cannot certify a remainder mismatch.

6 Translation and four-packet polarization

Fix the physical convention

$$n = s + m. \quad (8)$$

Then $q \mid n$ if and only if $m \equiv -s \pmod{q}$. Hence the moving hole is

$$\boxed{h_q = -s \pmod{q}.} \quad (9)$$

For a local packet $a_s(m) = a(s + m)$ and frequency v , write

$$z_{q,r}^{a,s}(v) = \sum_{m \equiv r \pmod{q}} a_s(m) e(vm/H).$$

The physical row differs by the common unit phase $e(vs/H)$, which cancels in every variance and absolute square. If the two polarized packets were first written at own origins s_β, s_w , with $d = s_w - s_\beta$, then the common-origin conversion is

$$W_r^{\text{common}}(v) = e(vd/H) W_{r-d}^{\text{own}}(v). \quad (10)$$

Both the relative residue shift and the phase are mandatory.

Let B_r, W_r be the common-origin residue rows for an ordered block pair (b, c) , and put

$$\bar{B} = \frac{1}{q} \sum_r B_r, \quad \bar{W} = \frac{1}{q} \sum_r W_r, \quad F_r = \sum_{m \equiv r \pmod{q}} \beta_b(m) \overline{w_c(m)}.$$

Define $a^{(j)} = \beta + i^j w$, $j = 0, 1, 2, 3$. The elementary complex polarization identity in the required orientation is

$$X\bar{Y} = \frac{1}{4} \sum_{j=0}^3 i^j |X + i^j Y|^2.$$

Applying it before any triangle inequality to Theorem 5.1 gives the exact ordered-pair moving-hole defect

$$\begin{aligned} M_{bc} = \int \psi(v) \sum_{q \in \mathcal{Q}} q \left\{ \frac{q}{q-1} \left[(B_0 - \bar{B})(\overline{W_0 - \bar{W}}) \right. \right. \\ \left. \left. - (B_{h_q} - \bar{B})(\overline{W_{h_q} - \bar{W}}) \right] \right. \\ \left. + \frac{q-2}{q-1} (F_{h_q} - F_0) \right\} dv. \end{aligned} \quad (11)$$

Equivalently, (11) is one quarter of the signed i^j -sum of the four scalar remainder defects. It is not obtained by four absolute-value estimates.

For later use, with $\hat{B}(k) = \sum_r B_r e_q(-kr)$, define $L_r = (B_r - \bar{B})(\overline{W_r - \bar{W}})$. Before the prefactor $q/(q-1)$ in (11), the bare leverage difference is

$$L_0 - L_{h_q} = \frac{1}{q^2} \sum_{k,l \neq 0} \hat{B}(k) \overline{\hat{W}(l)} (1 - e_q((k-l)h_q)). \quad (12)$$

Consequently the complete leverage correction inside the braces of (11) has coefficient $1/[q(q-1)]$ in front of the double sum. After the outer BDH weight q is included, its coefficient in M_{bc} is $1/(q-1)$. The selector vanishes when $k = l$. This off-equal-frequency cancellation belongs to the leverage term; the separate diagonal term $F_{h_q} - F_0$ does not disappear from (11).

7 Critical deterministic block bound

We now state a self-contained deterministic form of the block estimate. Let $\{I_b\}_{b \in \mathcal{B}}$ and $\{I_c\}_{c \in \mathcal{B}}$ be intervals of length at most $C_0 H$, with bounded overlap and ordered centers separated on the H scale. There are $J = |\mathcal{B}|$ effective blocks after discarding a Schwartz tail. Assume

$$|\beta_b(m)| \leq A_\beta \mathbf{1}_{I_b}(m), \quad |w_c(n)| \leq A_w \mathbf{1}_{I_c}(n).$$

The frequency integral is assumed to produce a kernel $K_H(m-n)$ satisfying, for every fixed $A > 2$,

$$|K_H(m-n)| \leq C_A (1 + |b-c|)^{-A} \quad (13)$$

whenever $m \in I_b, n \in I_c$. This normalization absorbs the fixed smooth cutoffs and is the one used in the V59 local scalar. For the cross-diagonal term we assume the usual bounded-overlap property: $I_b \cap I_c$ is empty unless $|b-c| \leq C_1$.

Theorem 7.1 (Moving-hole block payment). *Let $2 \leq Q \leq H$, and let $\mathcal{Q}$ be any set of integer moduli in $(Q, 2Q]$. Under the preceding hypotheses, the defects (11) satisfy*

$$\boxed{\sum_{b,c \in \mathcal{B}} |M_{bc}| \ll A_\beta A_w J (H^2 + HQ + Q^2),} \quad (14)$$

where the implied constant depends only on the support, overlap and Schwartz constants, not on the locations of the holes h_q .

Proof. For $r \in \mathbb{Z}/q\mathbb{Z}$ introduce the centered residue selector

$$\lambda_{q,r}(m) = \mathbf{1}_{m \equiv r \pmod{q}} - \frac{1}{q}.$$

On any interval of length at most $C_0 H$,

$$\sum_m |\lambda_{q,r}(m)| \ll \frac{H}{q} + 1, \quad (15)$$

uniformly in r . Indeed, the residue class occurs $O(H/q + 1)$ times and the constant part has total mass $O(H/q)$.

Integrate in v first. A leverage coordinate such as $(B_r - \bar{B})(\overline{W_r - \bar{W}})$ becomes a double coefficient sum with two centered selectors and kernel $K_H(m - n)$. Equations (13) and (15) therefore give, uniformly in the selected residue r ,

$$\left| \int \psi(v)(B_r - \bar{B})(\overline{W_r - \bar{W}}) dv \right| \ll A_\beta A_w (1 + |b - c|)^{-A} \left(\frac{H}{q} + 1 \right)^2.$$

The two leverage coordinates in (11) obey the same estimate. The outer q is retained. Summing the Schwartz separation first yields

$$\sum_{b,c} (1 + |b - c|)^{-A} \ll J,$$

not J^2 . Hence the total leverage contribution is

$$\begin{aligned} &\ll A_\beta A_w J \sum_{q \in \mathcal{Q}} q \left(\frac{H}{q} + 1 \right)^2 \\ &\ll A_\beta A_w J \left(H^2 \sum_{Q < q \leq 2Q} \frac{1}{q} + H \# \mathcal{Q} + \sum_{q \in \mathcal{Q}} q \right) \\ &\ll A_\beta A_w J (H^2 + HQ + Q^2). \end{aligned}$$

No prime number theorem is used; $\# \mathcal{Q} \leq Q$ suffices.

For the cross-diagonal, only $O(J)$ block pairs overlap. In each such pair, the residue support has size $O(H/q + 1)$, so its total contribution is

$$\ll A_\beta A_w J \sum_{q \in \mathcal{Q}} q \left(\frac{H}{q} + 1 \right) \ll A_\beta A_w J (HQ + Q^2),$$

which is already contained in (14). □

Remark 7.2. *The proof uses absolute values only after the exact four-packet polarization has produced the complete ordered-pair defect M_{bc} . It does not apply triangle inequalities separately to the four packets or destroy their algebraic cancellation.*

8 The V59 critical scale

At the frozen V59 scales

$$H = x^{21/32}, \quad Q = x^{1/3}, \quad J = \frac{x}{H} x^{o(1)},$$

we have $Q \leq H$ and

$$\begin{aligned} JH^2 &= xH x^{o(1)} = x^{53/32+o(1)}, \\ JHQ &= xQ x^{o(1)} = x^{4/3+o(1)}, \\ JQ^2 &= x^{97/96+o(1)}. \end{aligned}$$

The first term dominates, and

$$\frac{53}{32} = \frac{5}{3} - \frac{1}{96}. \quad (16)$$

Therefore Theorem 7.1 pays the moving-hole translation defect at the exact $1/96$ clock. The harmless $x^{o(1)}$ factor leaves the required strict margin for every fixed

$$\frac{1}{400} < \delta' < \frac{1}{96}. \quad (17)$$

This payment is local to the translation correction. It does not estimate the standard zero-hole remainder R_0 , and so it cannot be transferred to the full Gate-B scalar by itself.

9 Source boundary

Harper defines a gcd-grouped variance for general sequences and proves BDH asymptotics under explicit progressions, non-concentration, sparsity or integer-resemblance hypotheses [1]. For a prime modulus, the gcd- q group is the singleton zero class and has zero variance; thus the prime row is exactly the zero-hole variance V_0 . Theorem 3.1 supplies the exact crosswalk from this row to the translated physical row.

The source does not, however, prove the object still required by V59:

- the modulus subset is prime-only and carries outer weight q ;
- the variance is kernel-localized and has the exact $q - 2$ diagonal removed;
- four literal complex packets occur in one signed polarization;
- the required input hypotheses must hold uniformly in block and frequency; and
- the local estimates must be reassembled collectively without an internal modulus, packet, or block triangle loss.

Moreover, after exact diagonal subtraction the signed remainder is not positive, so an all-moduli estimate cannot simply be restricted to primes. We therefore source-lock the conclusion as follows: the Harper translation mismatch is no longer a fatal obstruction, but Harper is not a direct theorem attachment for the full Gate-B object.

10 Executable exact certificate

The accompanying standard-library artifact represents complex numbers as pairs of rational numbers and checks, without floating-point tolerances:

- all-hole identities for $q = 2, 3, 5, 7$;
- the exact diagonal lift and translation sign;
- four-packet polarization with weights $i^j/4$;
- the squared spectrum $q(q-2)/(q-1)^2$ and degeneracies;
- the corrected fixture (7);
- the rational exponent identity (16); and
- the explicit claim firewall.

An independent checker reimplements the arithmetic without importing the main implementation. Both normal and optimized Python modes are required to produce identical output. These finite checks are quality assurance and falsifiers for signs, factors, and fixtures; they are not evidence for the general theorems, whose proofs appear above.

11 Limitations and open theorem

The strongest positive result is the exact decomposition

$$\begin{aligned} \text{physical moving-hole remainder} &= \text{standard zero-hole remainder} \\ &\quad + \text{selected leverage defect} \\ &\quad + \text{selected energy defect,} \end{aligned}$$

together with a deterministic $x^{53/32+o(1)}$ payment for the complete translation defect. The strongest obstruction is equally explicit: the rank-two operator norm tends to one, so no saving is available from finite rank alone.

The first fatal remaining theorem is:

Prove a power-saving estimate for the standard zero-hole, prime-only, q -weighted, kernel-localized, exact-diagonal-subtracted signed remainder of the four literal V59 packets, uniformly over blocks and frequencies, and compile the block family collectively at a fixed saving greater than $1/400$.

The DFT selector $1 - e_q((k-l)h_q)$ in (12) is a reusable clue: it removes equal frequencies from the translation leverage term and may expose a post-dispersion or Kloosterman-cell interface. That arithmetic step remains open.

To conclude, TPC-207 is a proved structural L1 route advance. It proves the exact moving-hole projector and $q-2$ diagonal compiler and pays the translation subgate for every fixed $1/400 < \delta' < 1/96$. It proves no full Gate-B theorem, no global arithmetic advance, no L^2 estimate, no fixed-atom credit, and no twin-prime theorem.

References

- [1] Adam J. Harper. Simple Barban–Davenport–Halberstam type asymptotics for general sequences, 2024. Version 1.
- [2] Liang Wang. V59 polarized local BDH scalar compiler. Internal prime-dynamics-theory repository artifact, 2026. TPC_HANDOFF.md, V59.